

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Technology (B.Tech)

Semester – I / II (First Year) University Examination May 2014

B.Tech. (CE/EC/IT) / B.Tech. (CL/EE/ME)

CE101 Fundamentals of Computing & Programming (FOCP)

Date: 27/05/2014, Tuesday

Time: 01:30 p.m. to 4:30 p.m.

Maximum Marks:70

Instructions:

1. Attempt all questions.
2. Answers both sections in separate answer books.
3. Figure to the right indicates full marks.
4. Assume suitable data, if necessary and mention it.

SECTION – I

Q.1 Do as directed:

A. Write the statements and mention whether the following are True or False.

- a. `4Four` is an example of keyword.
- b. An integer value 2 occupies two bytes.
- c. `main()` is a special user define function in C.
- d. `sizeof` is an example of an operator & keyword in C.
- e. Two pointers can be added.

B. Classify between valid and invalid variable name in C.

(i) ROM	(ii) ANSI	(iii) Two_int
(iv) Float variable	(v) GJ@ELE	(vi) ID#22

Q.2 Answer the following questions.

1. Explain basic structure of a C program.
2. Explain: Various types of operators in C Programming language.

OR

Q.2 Answer the following questions.

1. What is flowchart? Explain various symbols used to draw a flowchart?
Draw a flowchart to print first N Fibonacci numbers.
2. Explain three classes of the data types. Explain primary data types in detail.

Q.3 Write programs in C.

1. Write a program to display following pattern using nested for loops.

```
* * * * *
* * * *
* * *
* *
*
```

2. Write a program to ADD two 3 X 3 Matrix. Read two matrices from the user.

OR

Q.3 Write programs in C.

1. Write a program to check whether the given number is Prime or not using do...while loop.
2. Write a C program to read 10 numbers from user and store them in an Array. Display Sum, Minimum and Average of the user entered numbers.

SECTION – II

- Q.4 Do as directed.** 11
1. What is the use of ternary operator? 02
 2. State the advantages and disadvantages of pointers. 02
 3. Compare structure with union. 02
 4. $(101)_{10} = (\quad)_{2}$ 02
 5. What is string? What is the use of `strlen()`? 02
 6. Narrate the use of `getch()`. 01
- Q.5 Answer the following questions.** 12
1. What is function? Explain the function definition, function prototype and function call with example.
 2. What is file management? List the different file management functions and explain the various file modes.
- OR
- Q.5 Answer the following questions.**
1. What is structure? How can we access structure members? Clear the use of nested structure using example.
 2. Explain: Two dimensional array with an example
- Q.6 Write programs in C.** 12
1. Define a structure named "Student" with rollno, name and percentage as members. Read data of 5 students from user and display them in proper format.
 2. Write a program to display the contents of a given file.
- OR
- Q.6 Write programs in C.**
1. Define a structure called CRICKET that will describe: player name, team average and batting average. Using CRICKET, declare an array player with 5 elements and write a program to read the information about all the 5 players and print team-wise list containing names of players with their batting average.
 2. Write a program to do swapping of two elements using function with two pointers as arguments.
